Status	Finished
Started	Thursday, 22 October 2026, 2:09 AM
Completed	Thursday, 22 October 2026, 2:11 AM
Duration	2 mins 34 secs
Grade	**12.00** out of 15.00 (**80**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is large and unsorted
- c. When the list is large and sorted
- d. When the list is small and sorted

Question **2**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(1)
- c. O🥐
- d. O(n log n)

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search stops
- c. The list is sorted
- d. The search continues from the beginning

Question **4**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n/2
c. n ✓
d. 1

Question **5**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94
b. 90 and 100 ✗
c. 94 and 99
d. 90 and 99

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search
b. Merge search
c. Linear search ✓
d. Quick search

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(1)
c. O(n log n)
d. O👎 ✓

Question **8**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)

b. Binary ✓

c. Linear

d. Hash

Question **9**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100

b. 1000

c. 10 ✓

d. 500

Question **10**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is somewhere in the middle of the array ✓

b. Item is the last element in the array or item is not there at all

c. When the item is the last element in the array

d. When the item is not the array at all

Question **11**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The search continues in the right sublist

c. The list is sorted

d. The search ends successfully ✓

Question **12**

Incorrect

Mark 0.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By comparing each element in the list sequentially
c. By sorting the list first ✗
d. By dividing the list into halves

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. In the entire list
c. At the beginning of the list
d. In the left sublist ✓

Question **14**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list
b. Divide the list into two equal parts
c. Sort the list ✗
d. Compare the target element with the first element in the list

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓
b. The list must be sorted
c. There should be direct access to the middle element in any sublist
d. Number values should only be present

Number values should only be present